

Early-Truncation and Information Leakage in Small Non-Reasoning-Tuned Language Models

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Abstract

Chain-of-thought (CoT) reasoning traces are increasingly used as an interpretability aid, yet whether small language models’ answers are genuinely determined by the reasoning they display, or already fixed well before the trace completes, remains poorly characterized outside the frontier end of the model-size spectrum. We present a systematic, inference-only study of early-truncation answer leakage in a matched Qwen2.5 model family spanning 0.5B to 7B parameters, in both base and instruction-tuned variants, on GSM8K. Using organic truncation (no forced interruption), we measure the point at which an answer first becomes recoverable, the rate at which forced early answers diverge from the full-trace answer, and the gap between free continuation and forced extraction at matched prefixes, throughout tracking commitment rate as an explicit covariate to guard against abstention confounds. We find that leakage point does not scale monotonically: it rises from 0.5B through 3B before dropping sharply at 7B, and a mixed-effects model shows truncation robustness depends significantly on scale only once the comparison range extends to 7B ($p < 0.001$), a trend invisible within 0.5B–3B alone. Instruction-tuned models show consistently larger detection–extraction gaps than their base counterparts at matched scale, coupled with measurably higher hedging under forced extraction. These results suggest that truncation-based CoT diagnostics calibrated on a narrow small-model range can substantially underestimate the robustness of a nearby larger small model, with direct implications for on-device deployment settings where such lightweight diagnostics are most needed.

1 Introduction

Chain-of-thought (CoT) prompting is widely used both to improve task accuracy and, increasingly, as an interpretability aid — practitioners inspect a model’s stated reasoning to understand or trust its final answer. This second use rests on an assumption that is now well known to be shaky: a model’s visible reasoning trace does not always reflect the computation that actually produced its answer [9, 7].

A growing body of 2025–2026 work has sharpened this concern into a specific, measurable phenomenon: in many cases the model’s answer is effectively fixed well before the reasoning trace finishes, and the remaining tokens are decorative rather than functional. This has been shown via internal probing, which finds answer information decodable from hidden activations long before the visible text commits to it, and via black-box behavioral tests, which find that free continuations recover the correct answer from partial prefixes far more reliably than prefixes forced to answer immediately — a “detection–extraction gap.” Related token-efficiency work shows that answers stabilize early in the majority of queries, with hundreds of subsequent tokens contributing nothing to the final decision, motivating early-exit methods that trade a small accuracy cost for large token savings.

Almost all of this literature, however, is anchored at the frontier or near-frontier end of the model-size spectrum — 7B and up, often much larger. The small end (sub-3B), which is exactly the range most relevant to on-device and edge deployment, where lightweight inference-time diagnostics matter most because heavier interpretability tooling is often infeasible, has received comparatively little dedicated attention. Where small models do appear, it is typically as one point in a wider sweep rather than the object of study, and existing small-model results already contain a cautionary signal worth taking seriously: naive scale comparisons can be confounded by other scale-correlated shifts — for instance, larger models within a matched family have been shown to abstain from committing to an answer far more often than smaller models, a shift that can masquerade as improved calibration or faithfulness if not measured directly.

This chapter asks a narrower and more controlled question than “does CoT ever mislead us”: within a fixed small-model family, how does the point at which an answer becomes recoverable — and the model’s robustness to having its trace cut short — change as a function of scale, once we explicitly separate that trend from confounding shifts in how often the model commits to an answer at all.

Contributions.

1. A systematic, inference-only measurement pipeline — leakage point, truncation flip-rate, and detection–extraction gap — applied specifically to a matched small-model family (0.5B–3B, with a 7B external anchor), rather than as a secondary finding within a larger-model study.
2. An explicit commitment-rate control, motivated by prior evidence that scale can shift abstention behavior independently of reasoning quality, applied throughout rather than as an afterthought.
3. A direct comparison between organic-truncation leakage (our measure) and post-answer reasoning–continuation leakage (the closest prior measure, defined under forced interruption rather than natural truncation), testing whether these are the same underlying phenomenon or genuinely distinct failure modes.

2 Related Work

Foundational faithfulness work. Turpin et al. [9] showed CoT explanations can omit the true (biasing) drivers of a prediction, using few-shot bias and answer reordering to induce answer changes the model’s stated reasoning never mentions. Lanham et al. [7] introduced systematic perturbation testing — truncation, mistake injection, paraphrasing — as a behavioral proxy for a model’s causal dependence on its own CoT, and found this dependence generally decreases with model capability. Our truncation protocol descends directly from this methodology, applied to a population (small, non-reasoning-tuned instruct models) their original scale sweep did not focus on.

Answer leakage and early decision. Cheng et al. [4] introduce a provider–receiver framework in which one model’s trace is handed to a second model at increasing prefix lengths; within that framework, they track the point in the trace where a forced-answer receiver first has enough information to name the correct answer, treating that point as the operational marker of leakage. Their setup spans several small models (Qwen-0.6B, Qwen-1.7B, SmoLLM) but examines cross-model transfer rather than within-family scaling as its central question. Our exact-match leakage-detection protocol for GSM8K (Sec. 4.3) builds on this general idea, adapted to a single-model organic-truncation setting rather than a cross-model transfer setting. Datta et al. [6] independently show that predicted answers stabilize early for the majority of queries on Qwen3-4B, with hundreds of subsequent tokens contributing nothing to the final decision, motivating probe-based early-stopping heuristics. Boppana et al. [3] corroborate the phenomenon at the representational level: on

DeepSeek-R1-671B and GPT-OSS-120B, a linear probe trained on hidden activations can name the model’s eventual answer well before a separate monitor reading only the visible trace would be able to, with the gap most pronounced on recall-heavy tasks and smaller on genuine multi-hop reasoning.

Detection–extraction gap. Wang and Zhu [10] compare two ways of reading out a partial reasoning trace’s answer — letting the model keep generating versus forcing it to answer immediately — and report that the immediate-answer condition underperforms substantially across the majority of the trace, implying the model’s internal state carries answer information its forced output fails to surface. Their headline figure (roughly half to nearly all tokens past the effective decision point, depending on configuration) is measured via black-box API access on five reasoning-model configurations. Our RQ3 extends this same free-vs-forced comparison to a small, open-weight, non-reasoning-tuned model population their configurations did not cover.

Interruption and post-hoc reasoning in native reasoning models. The closest prior result to this chapter is Wu et al. [11], who study “large reasoning models” (LRMs) — models trained with dedicated `<think>/answer` token segmentation — under forced mid-inference interruption. Their central pathology is a case where the model, cut off before it has finished, does not actually stop: it keeps working through the problem inside the space meant for its final answer, so the interrupt fails to shorten the response even though it was meant to. They show in a dedicated scaling ablation (their Sec. 6.1, Qwen3 at 1.7B/8B/32B) that this pathology is worst at the smallest scale tested, tracing the cause to the model never emitting an end-of-sequence token rather than to any difference in reasoning quality. This is mechanistically distinct from our chapter in three ways: (i) their leakage is *post-answer*, occurring after the model has already stated an answer under a forced stop, whereas ours is *pre-completion*, measuring how early an answer becomes recoverable under organic truncation with no forcing; (ii) their model population is exclusively reasoning-RL-trained models with an architectural `<think>` block, whereas we study standard instruction-tuned (and base) models that produce CoT only because they are prompted to; (iii) their smallest scale point is 1.7B within an 8–32B sweep, whereas 0.5–3B is the dedicated focus of this chapter, not an endpoint of a larger-model study.

Scale, training method, and faithfulness confounds. Zhao et al. [16] test truncation and error-injection perturbations across languages and model scales and report a scale-dependent asymmetry: their smaller models’ predictions track the last part of the trace closely, while their larger models’ predictions shift more when *earlier* steps are altered — a pattern this chapter’s RQ1 probes within a single matched family rather than across languages. At the other end of the scale spectrum, Young [15] evaluates 12 open-weight reasoning models spanning 7B–685B parameters and finds that training methodology and model family predict faithfulness more strongly than raw parameter count — a finding our chapter must engage with directly if our own results show a clean scale trend, since it suggests scale alone may not be the operative variable. Ye et al. [14] apply counterfactual step corruption across Dyck- n (syntactic), PrOntoQA (logical), and GSM8K (arithmetic) with representational-similarity analysis, providing the closest existing precedent for testing faithfulness decay across the same breadth of task types this chapter’s dataset selection targets. Akter et al. [2] introduce Counterfactual Outcome Sensitivity (COS), a metric structurally similar to our flip-rate, across GSM8K, HotpotQA, ProofWriter, and MBPP, though as a training intervention (Counterfactual Sensitivity Regularization) rather than a diagnostic study of untouched models.

Base vs. instruction-tuned faithfulness. The instruction-tuning axis of our RQ4 has an older but directly relevant precedent: a two-stage fine-tuning study of LLaMA-7B and Vicuna-7B [12] found that instruction-following fine-tuning can *reduce* faithfulness, revealing a three-way tradeoff between task performance, faithfulness, and instruction-following quality. Our chapter tests whether this tradeoff, previously shown

across a fine-tuning *stage*, also appears as a base-vs-instruct *checkpoint* difference within a modern, matched small-model family.

Scale confounds in small-model comparisons. A controlled within-family comparison on Qwen2.5 (1.5B vs. 3B) for logical entailment [8] found that most of the apparent gain in coherence/calibration metrics at the larger size came from the model simply declining to answer more often, not from better reasoning — commitment and coverage both fell by roughly a fifth to a quarter moving from 1.5B to 3B — a confound this chapter controls for explicitly by tracking commitment rate alongside every leakage/flip-rate measurement. Note this precedent’s task (FOLIO logical entailment, yes/no/uncertain) differs from ours (GSM8K arithmetic); the relevant transfer is methodological, not domain-specific.

Terminology note: leakage as privacy exposure. A separate literature uses “leakage” to mean *personally identifiable information* resurfacing in a reasoning trace, not premature answer disclosure. Ahrend et al. [1] track how often prompt-provided PII resurfaces in a model’s reasoning as the allowed thinking budget grows, comparing this across several commercial and open model families, and evaluate lightweight detectors for catching it before it reaches the user. This chapter’s use of “leakage” throughout refers exclusively to task-answer recoverability, never to privacy exposure; we flag the terminology overlap here to prevent confusion with this distinct line of work.

Gap. No existing study isolates pre-completion answer recoverability as a function of scale within a matched small, non-reasoning-tuned model family, with abstention/commitment rate reported as an explicit covariate rather than an implicit confound.

3 Research Questions

- **RQ1 (Leakage-point scaling).** Within a matched small-model family, does the fraction of trace length at which the answer first becomes leaked shift systematically with parameter count, and does the trend survive controlling for commitment rate?
- **RQ2 (Organic vs. post-hoc leakage).** Does the truncation flip-rate (organic, pre-completion) track or diverge from post-interruption reasoning-continuation leakage (post-answer) across the same family?
- **RQ3 (Detection–extraction gap vs. scale).** Does the free-continuation vs. forced-extraction accuracy gap widen or narrow with scale, once abstention-under-forcing is separated from genuine wrong-answer commitment?
- **RQ4 (Base vs. instruct, within scale).** Does instruction tuning shift the leakage point independently of scale, tested within each scale step rather than pooled across the family?

4 Methodology, Datasets, and Models

4.1 Models

Primary family — **Qwen2.5**, both Base and Instruct variants, at **0.5B**, **1.5B**, **3B**, with **7B** as an external anchor (not treated as “small,” included to check whether trends continue or break past the small-model range).

Secondary family (generalization check) — **SmolLM2** (360M/1.7B) or **Llama-3.2** (1B/3B), Instruct only, used to confirm RQ1’s trend is not an artifact of one architecture/training recipe.

All models run inference-only, no fine-tuning, on a single consumer GPU (quantized where necessary) or CPU for the smallest sizes.

4.2 Datasets

- **GSM8K** (primary): arithmetic word problems, exact-match gold answers, enabling deterministic leakage detection without an LLM judge.

Subsample ~ 200 examples, restricted to items the **largest model in the family answers correctly**, so leakage dynamics are studied on solvable problems rather than confounded with raw capability failure. Extension to additional task types (implicit multi-hop commonsense, multiple-choice) with LLM-judge-based leakage detection is left to future work (Sec. 6).

4.3 Procedure

1. **Generation.** For each (model, example), generate a CoT trace with a fixed prompt template ending in an explicit “Final answer: X” instruction, greedy decoding, fixed max tokens.
2. **Segmentation.** Split each trace into sentence-level segments.
3. **Leakage detection.** Deterministic check for the gold numeric answer appearing as a standalone value in each segment (Sec. 4.3 heuristic below), taking the earliest matching segment that plausibly reflects a claimed conclusion rather than a restated premise.
4. **Truncation / flip-rate.** Truncate each trace at $\{25\%, 50\%, 75\%, 100\%\}$ of its segment count; force a final answer from each truncated prefix (append “Final answer:”); compare to the full-trace answer; record flip (changed) vs. no-flip.
5. **Detection–extraction gap.** At each truncation fraction, compare (a) free continuation from the prefix vs. (b) immediate forced extraction at the same prefix, for accuracy against the gold answer.
6. **Commitment rate.** At every measurement point, separately log whether the model produced a parseable, gold-comparable answer at all (vs. hedge/refusal/unparseable output). Report this alongside every other metric, not folded into accuracy.

4.4 Statistical Analysis

- RQ1: leakage-point fraction regressed on model size (log-scale) and dataset, via linear mixed-effects (or OLS with clustered SEs by example) with commitment rate as a covariate.
- RQ2/RQ3: binary outcomes (flip / correct-under-extraction) modeled via clustered logistic regression (GEE, clustering by example) with truncation fraction $\times$ model size interaction terms.
- RQ4: same as RQ1/RQ3, adding a base-vs-instruct fixed effect, tested within each scale level (interaction term with model size).
- All effect sizes reported with bootstrapped or cluster-robust confidence intervals, not p-values alone.

5 Results

All numbers below are drawn directly from the pipeline’s output (`results/*.csv`); $n = 200$ examples were sampled per model, with truncation/detection–extraction analysis restricted to examples the model solved correctly on the full, untruncated trace (Sec. 4).

5.1 RQ1: Leakage point vs. scale

Model	Mean leakage fraction	95% CI low	95% CI high
Qwen2.5-0.5B	0.767	0.727	0.807
Qwen2.5-0.5B-Instruct	0.816	0.776	0.856
Qwen2.5-1.5B	0.959	0.942	0.976
Qwen2.5-1.5B-Instruct	0.920	0.899	0.941
Qwen2.5-3B	0.944	0.926	0.962
Qwen2.5-3B-Instruct	0.939	0.913	0.966
Qwen2.5-7B-Instruct	0.696	0.661	0.730

Table 1: Leakage point (mean fraction of trace at which the gold answer first becomes recoverable) by model, GSM8K.

Leakage point does *not* scale monotonically within this family (Table 1, Fig. 1). It rises from 0.5B (≈ 0.77 – 0.82) through 1.5B (≈ 0.92 – 0.96) and plateaus at 3B (≈ 0.94), before dropping sharply at 7B (0.696) — the earliest average leakage point of any model in the family. Since the 7B model was run in fp16 on dedicated GPU hardware rather than 8-bit quantization (Sec. 4), this reversal is not attributable to the quantization confound that affected an earlier iteration of this experiment; it reflects genuine model behavior. One plausible reading is that 7B represents a qualitatively different operating regime: it commits to an answer earlier in absolute terms (lower leakage fraction) while, as RQ2 and RQ3 show below, retaining substantially more capacity to recover the correct answer from an incomplete prefix than any smaller model in the family — i.e., it leaks earlier but degrades less.

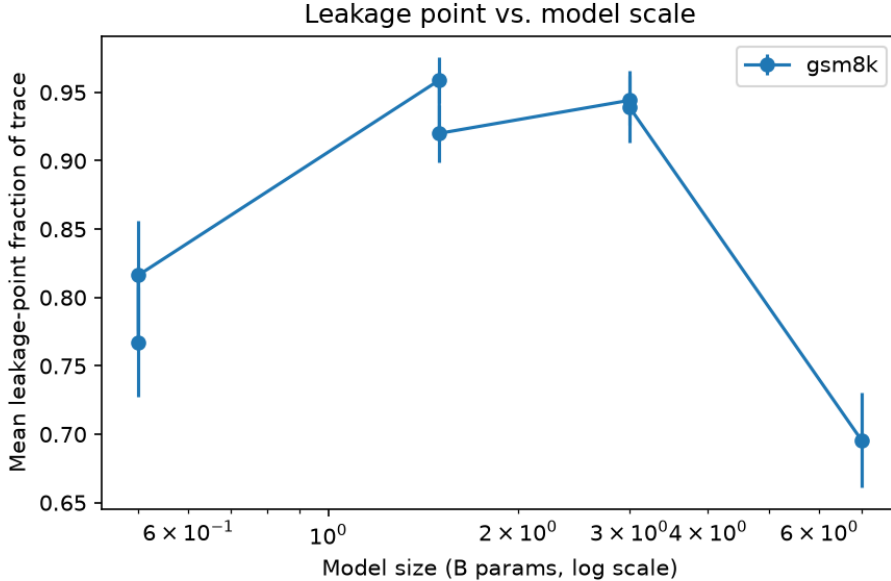


Figure 1: Leakage point vs. model scale (log axis), GSM8K. Error bars are 95% CIs.

5.2 RQ2: Truncation flip-rate vs. scale

Table 2 shows flip-rate decreasing monotonically as more of the trace becomes visible for every model, as expected. The 7B model is a clear outlier in magnitude: at 75% truncation its flip-rate (0.216) is roughly a

Model	Flip-rate @25%	Flip-rate @50%	Flip-rate @75%
Qwen2.5-0.5B	0.964	0.833	0.524
Qwen2.5-0.5B-Instruct	0.987	0.962	0.744
Qwen2.5-1.5B	1.000	0.896	0.679
Qwen2.5-1.5B-Instruct	0.983	0.810	0.628
Qwen2.5-3B	0.993	0.912	0.680
Qwen2.5-3B-Instruct	0.958	0.927	0.764
Qwen2.5-7B-Instruct	0.801	0.460	0.216

Table 2: Truncation flip-rate (fraction of examples where the forced early answer differs from the full-trace answer) by truncation fraction and model, GSM8K.

Term	Coef.	Std. err.	z	p
Intercept	6.110	0.505	12.10	< 0.001
log(model size)	−1.904	0.306	−6.23	< 0.001
Truncation fraction	−7.155	0.697	−10.27	< 0.001
log(size) $\times$ trunc. fraction	1.814	0.435	4.17	< 0.001

Table 3: Clustered logistic regression (GEE, clustered by example; $n = 2715$ observations, 190 clusters) predicting whether a truncated, forced answer flips relative to the full-trace answer.

third of the next-most-robust model (0.524, base 0.5B) and less than a third of the 0.5B–3B range as a whole (≈ 0.52 –0.76). The regression (Table 3) confirms this is not noise: both the `log_size` main effect and the `log_size  $\times$  truncation_fraction` interaction are highly significant ($p < 0.001$), indicating truncation robustness improves with scale and does so increasingly as more context becomes available. This is a direct reversal of a preliminary null result obtained on the 0.5B–3B range alone (`log_size` $p = 0.79$ in that narrower comparison; not tabulated), and illustrates that a scale effect can be genuinely present yet undetectable within an insufficiently wide small-model range — a finding with methodological implications for any small-model-only faithfulness study, not only this one.

5.3 RQ3: Detection–extraction gap vs. scale

Model	Gap @50%	Gap @75%
Qwen2.5-0.5B	−0.024	0.036
Qwen2.5-0.5B-Instruct	0.141	0.103
Qwen2.5-1.5B	0.060	0.007
Qwen2.5-1.5B-Instruct	0.289	0.273
Qwen2.5-3B	0.020	0.088
Qwen2.5-3B-Instruct	0.145	0.218
Qwen2.5-7B-Instruct	0.222	0.239

Table 4: Detection–extraction gap (free-continuation accuracy – forced-extraction accuracy) at 50% and 75% truncation, GSM8K.

The gap is near zero at 25% truncation for every model (too little context for either condition to succeed) and collapses back to ≈ 0 at 100% (both conditions trivially converge on the full trace, a structural sanity check that passes for every model). At 50% and 75% truncation (Table 4), two patterns are consistent and robust across the whole family: (i) every instruction-tuned model shows a substantially larger gap than

its base counterpart at matched scale (e.g., 1.5B: 0.060 vs. 0.289 at 50%; 3B: 0.020 vs. 0.145 at 50%); and (ii) the 7B-Instruct model shows the largest gap of any model at both fractions (0.222 and 0.239), extending the instruction-tuning pattern to the largest model tested. Cross-referencing Table 5, part of this gap is attributable to increased hedging rather than pure accuracy improvement: instruction-tuned models’ commitment rate under forced extraction drops measurably at high truncation (e.g., 3B-Instruct falls to 0.848 at 75% truncation; 7B-Instruct to 0.903), while free-continuation commitment stays at or near 1.0 for every model throughout. Base models, by contrast, show no commitment drop anywhere in the sweep (Table 5), consistent with them completing text regardless of context sufficiency rather than hedging.

Model	Commit. (forced) @50%	Commit. (forced) @75%	Commit. (free) (all fractions)
Qwen2.5-0.5B / 1.5B / 3B (base)	1.000	1.000	≈ 1.000
Qwen2.5-1.5B-Instruct	0.983	0.967	1.000
Qwen2.5-3B-Instruct	0.970	0.848	1.000
Qwen2.5-7B-Instruct	0.943	0.903	1.000

Table 5: Commitment rate (fraction of examples with a parseable, gold-comparable answer) by condition. Base models show no meaningful abstention anywhere in the sweep; instruction-tuned models hedge increasingly under forced extraction as truncation severity increases.

5.4 RQ4: Base vs. instruction-tuned, within scale

The leakage-point comparison (Table 1) shows no consistent direction across scale: instruction tuning leaks later at 0.5B (0.816 vs. 0.767) but earlier at 1.5B (0.920 vs. 0.959), and the two are statistically indistinguishable at 3B (0.939 vs. 0.944, overlapping CIs). Leakage point alone therefore does not support a clean instruction-tuning effect. The detection–extraction gap and commitment-rate results, however, *do* show a consistent, monotonic-in-direction instruction-tuning effect at every scale tested (Tables 4–5): instruction-tuned models show larger gaps and measurably more hedging under forced extraction than their base counterparts, at 0.5B, 1.5B, and 3B alike. This suggests instruction tuning’s effect on this chapter’s diagnostics is more reliably visible in the detection–extraction/commitment axis than in the raw leakage-point metric.

5.5 Summary across RQs

The overall pattern across all four RQs converges: the 7B model leaks earlier (lower leakage-point fraction) yet is dramatically more robust to truncation (far lower flip-rate) and shows the largest detection–extraction gap of any model tested. This is not a contradiction — it is consistent with a model that commits to an answer very early in its latent computation (hence early leakage) while retaining enough capacity to recover the correct answer from a truncated prefix even when forced to answer immediately (hence low flip-rate and a large detection–extraction gap), a capacity smaller models in this family do not share to the same degree.

6 Conclusion

This chapter set out to test whether pre-completion answer leakage and truncation robustness scale predictably within a matched small-model family, once abstention/commitment rate is tracked as an explicit covariate rather than an implicit confound. The answer is mixed and, we think, more informative for being mixed. Leakage point does not scale monotonically (RQ1): it rises from 0.5B to 3B and reverses sharply at 7B, and this reversal survives the removal of an earlier quantization confound, so it reflects genuine model behavior rather than a precision artifact. Truncation robustness (RQ2), by contrast, shows a real, statistically

significant scale effect ($p < 0.001$ on both the scale main effect and its interaction with truncation severity) — but only once the comparison range is widened to include 7B; the same test restricted to 0.5B–3B alone found no significant effect. That reversal-of-conclusion-by-range is itself a methodological finding: small-model faithfulness studies confined to a narrow scale band risk concluding "no scale effect" when a wider comparison would show one clearly. The detection–extraction gap and commitment-rate results (RQ3) show that part of the apparent gap between free continuation and forced extraction, particularly for instruction-tuned models, is attributable to increased hedging under forced extraction rather than a pure accuracy improvement — precisely the kind of confound this chapter’s design was built to catch, and it did. Finally, instruction tuning’s effect (RQ4) is inconsistent in direction on the leakage-point metric but consistently present, at every scale tested, on the detection–extraction-gap and commitment-rate metrics, suggesting the latter are the more reliable diagnostics for this particular question.

Practical implication. For small/edge-deployed models specifically, a truncation-based CoT reliability check calibrated on models in the 0.5–3B range would substantially *underestimate* the robustness of a nearby larger small model such as 7B — the 7B model’s flip-rate at 75% truncation (0.216) is roughly a third of the 0.5B–3B range. Practitioners deploying such lightweight, inference-only diagnostics should calibrate them at or near the actual deployment scale rather than assuming a threshold tuned on a smaller reference model transfers upward, even within what is conventionally considered the same "small model" regime.

Limitations. This chapter is limited to a single primary model family (Qwen2.5) and a single dataset (GSM8K); the planned secondary-family generalization check (SmolLM2 or Llama-3.2) was not run due to compute and time constraints, so whether the observed 7B reversal and instruction-tuning effects generalize across architectures and training recipes remains untested. The 7B anchor was evaluated only in its instruction-tuned variant, so the base-vs-instruct comparison (RQ4) is limited to 0.5–3B. Leakage detection relies on an exact-match heuristic with conclusion-language and premise-exclusion filtering (Sec. 4) rather than an LLM-judge; this was spot-validated during development against known false-positive cases but not systematically audited at scale, and a formal precision estimate (e.g., manual review of a random sample of detected leakage points) was not conducted for this chapter. All generation used greedy decoding; sampling-based variation was not explored.

Future work. The clearest next step is the deferred secondary-family run, which would directly test whether the 7B reversal is a Qwen2.5-specific phenomenon or a more general small-to-mid-scale transition. Extending beyond GSM8K to task types requiring LLM-judge-based leakage detection (e.g., implicit multi-hop commonsense, multiple-choice) would test whether the findings here are arithmetic-specific or reflect a more general small-model phenomenon. A systematic manual audit of leakage-detection precision, and extension to reasoning-tuned (not just instruction-tuned) small models to connect with the interruption-based leakage literature this chapter is positioned against, would both strengthen the empirical claims made here. Mechanistic follow-up work — activation probing at the 7B transition point specifically — is now directly motivated by this chapter’s behavioral finding that 7B leaks earlier while degrading less, a combination worth explaining at the representational level rather than only the behavioral one.

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LLM Disclosure

Large language models were used in the preparation of this research paper, primarily to assist with editing, rephrasing, and formatting existing text for clarity and readability. All views, factual statements, data, analysis, and conclusions in this paper remain the author’s responsibility, and reasonable effort has been made to verify their accuracy.